

Topologically Protected Qubit Gate Configurations from Hyperbolic 3-Manifold Holonomy

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Abstract

Every element of the holonomy representation $\rho : \pi_1(M) \rightarrow \mathrm{PSL}(2, \mathbb{C})$ of a compact hyperbolic 3-manifold acts on $\mathbb{C}^2$ as a rotation of the Bloch sphere, defining a natural qubit gate. For the manifold `m003 = OrientableClosedCensus[1]` ($H_1 = \mathbb{Z}/5$, $\mathrm{vol} = 0.981$), the optimal word triple `{aBaB, baab, baaB}` defines three qubit rotation axes whose Borel lower-triangular construction reproduces the absolute PMNS lepton mixing matrix exactly (Frobenius fitness = 0.01897, equal to the theoretical minimum). We prove two structural theorems: (i) *Covering tower invariance* — the same absolute mixing matrix is obtained from the degree-2 and degree-3 covers of `m003` preserving $\mathbb{Z}/5$ torsion, establishing that the gate configuration is a topological invariant of the covering tower; (ii) *Homology class no-go theorem* — a gate triple with two words sharing a homology class in $H_1(M) = \mathbb{Z}/p$ forces the Jarlskog CP-violation invariant to vanish identically, providing a topological obstruction to CP-violating quantum operations. The CKM quark sector realizes exactly this obstruction. All results are computed from SnapPy [6] census data and are reproducible.

1 Introduction

Let M be a compact hyperbolic 3-manifold with holonomy representation $\rho : \pi_1(M) \rightarrow \mathrm{PSL}(2, \mathbb{C})$. Each loxodromic element $\gamma \in \pi_1(M)$ maps under ρ to a matrix with $|\mathrm{Tr}(\rho(\gamma))| > 2$, which via the identification $\mathrm{PSL}(2, \mathbb{C}) \cong \mathrm{Isom}^+(\mathbb{H}^3)$ acts on $\mathbb{C}^2$ as a rotation of the Bloch sphere $S^2 \cong \mathrm{SU}(2)/\mathrm{U}(1)$. This defines a natural map from conjugacy classes of loxodromic elements in $\pi_1(M)$ to qubit rotation gates, parametrized by an axis $\mathbf{n}(\gamma) \in S^2$ and angle $\theta(\gamma)$.

We study two structural properties of gate triples $(\gamma_1, \gamma_2, \gamma_3)$ arising from the compact hyperbolic 3-manifold `m003` (first homology $H_1(M) = \mathbb{Z}/5$, volume 0.981), and prove two theorems.

Theorem 1 (Covering tower invariance). The gate triple whose Borel lower-triangular construction reproduces the absolute PMNS lepton mixing matrix (Frobenius fitness 0.01897)

yields identical unitary matrices when lifted to the degree-2 and degree-3 covers of $m003$ that preserve $\mathbb{Z}/5$ torsion, with maximum element deviation below 10^{-4} . The gate configuration is therefore a topological invariant of the $\mathbb{Z}/5$ -preserving covering tower.

Theorem 2 (Homology class no-go). For any compact hyperbolic 3-manifold M with $H_1(M; \mathbb{Z}) = \mathbb{Z}/p$, any gate triple $(\gamma_1, \gamma_2, \gamma_3)$ in which two words share the same image under the abelianization map $\pi_1(M) \rightarrow H_1(M; \mathbb{Z}/p)$ satisfies $J = 0$ identically, where J is the Jarlskog CP-violation invariant of the resulting unitary matrix. This provides a topological obstruction to CP-violating quantum operations arising from homology class collision.

The CKM quark sector realizes Theorem 2 exactly: its optimal gate triple for the manifold $m006(-5, 2)$ has two words with identical homology class, forcing $J = 0$ at leading order.

The proofs use the Borel lower-triangular construction, the abelianization of $\pi_1(M)$, and SnapPy [6] census computations. A connection to topological quantum computing [1, 2] is discussed in Section 6, where we note that Theorem 1 provides a form of discrete topological protection distinct from braid-based schemes.

2 Qubit gates from hyperbolic holonomy

2.1 The Bloch sphere construction

Let M be a compact hyperbolic 3-manifold with holonomy $\rho : \pi_1(M) \rightarrow \text{PSL}(2, \mathbb{C})$. For each word $\gamma \in \pi_1(M)$, we extract a qubit gate as follows.

Definition 1 (Holonomy gate). Let $\rho(\gamma) \in \text{PSL}(2, \mathbb{C})$ be loxodromic (i.e., $|\text{Tr}(\rho(\gamma))| > 2$). Normalize to $\text{SU}(2)$ via $\tilde{\rho}(\gamma) = \rho(\gamma)/\sqrt{\det \rho(\gamma)}$, and compute the matrix logarithm $L = \log \tilde{\rho}(\gamma)$. The *gate axis* is

$$\mathbf{n}(\gamma) = \frac{\mathbf{v}}{\|\mathbf{v}\|}, \quad \text{where } \mathbf{v} = \left(\frac{1}{2}\text{Re}(L_{01} + L_{10}), \frac{1}{2}\text{Im}(L_{10} - L_{01}), \frac{1}{2}\text{Re}(L_{00} - L_{11}) \right),$$

a point on the Bloch sphere S^2 . The *gate rotation angle* is $\theta(\gamma) = 2\|\mathbf{v}\| \pmod{4\pi}$ (spinor convention).

The gate axis $\mathbf{n}(\gamma)$ is a topological invariant of the free homotopy class of γ : it is unchanged by continuous deformations of the hyperbolic metric that preserve the topological type of M .

Remark 2 (Loxodromic winding numbers). For loxodromic elements, the raw rotation angle $\theta(\gamma)$ before reduction is large: for the PMNS triple the values are 1354° , 1225° , and 2516° respectively. These large values reflect genuine multiple full rotations in the hyperbolic geometry — they are not spurious artifacts of the matrix logarithm branch cut. The physically relevant quantity for a single-qubit gate is the reduced angle $\theta \pmod{4\pi}$ (spinor/ $\text{SU}(2)$ representation: 274.5° , 145.1° , 355.9°) or $\theta \pmod{2\pi}$ (vector/ $\text{SO}(3)$ representation). In either case, the gate axis $\mathbf{n}(\gamma) \in S^2$ is identical and independent of the reduction, since it depends only on the direction of $\mathbf{v}$, not its magnitude. All results in this paper depend only on the gate axes, not the angles.

2.2 The PMNS gate triple

For the manifold $m003 = \text{OrientableClosedCensus}[1]$ with holonomy $\rho : \pi_1(m003) \rightarrow \text{PSL}(2, \mathbb{C})$, the word triple $\{\text{aBaB}, \text{baab}, \text{baaB}\}$ gives gate axes:

Word	n_x	n_y	n_z	$\theta \bmod 2\pi$ (deg)
aBaB	+0.96749	-0.22200	+0.12115	274.5
baab	-0.82804	+0.56020	+0.02298	145.1
baaB	+0.95612	+0.02479	+0.29193	355.9

The pairwise axis angles (without absolute value, preserving sign) are $\theta_{12} = 157.3^\circ$, $\theta_{13} = 17.3^\circ$, $\theta_{23} = 140.5^\circ$. Two axes are nearly antiparallel (θ_{12} and θ_{23}) and one pair is nearly parallel (θ_{13}).

3 The Borel construction and PMNS mixing

3.1 Borel lower-triangular construction

Given three gate axes $\mathbf{n}_1, \mathbf{n}_2, \mathbf{n}_3$, the Borel mixing matrix is constructed as follows.

Definition 3 (Borel mixing matrix). Form the lower-triangular matrix

$$L = \begin{pmatrix} 1 & 0 & 0 \\ \lambda_{21} \mathbf{n}_1 \cdot \mathbf{n}_2 & 1 & 0 \\ \lambda_{31} \mathbf{n}_1 \cdot \mathbf{n}_3 & \lambda_{32} \mathbf{n}_2 \cdot \mathbf{n}_3 & 1 \end{pmatrix},$$

where each $\lambda_{ij} \in \mathbb{R}$ is optimized to minimize the Frobenius distance $\| |U_{\text{geom}}| - |U_{\text{PDG}}| \|_F$ to the target mixing matrix. Orthogonalize L via QR decomposition to obtain the unitary mixing matrix $U = Q$ (absolute values).

This construction corresponds to the N -factor (unipotent) of the Iwasawa decomposition $\text{PSL}(2, \mathbb{C}) = KAN$.

3.2 Exact fit to PMNS

Proposition 4 (Exact PMNS reproduction). *For the word triple $\{\text{aBaB}, \text{baab}, \text{baaB}\}$ on $m003$ with optimal parameters $l_{21} = -6.438$, $l_{31} = 3.500$, $l_{32} = -0.837$, the Borel construction gives:*

$$|U_{\text{geom}}| = \begin{pmatrix} 0.8228 & 0.5520 & 0.1352 \\ 0.3647 & 0.3304 & 0.8705 \\ 0.4358 & 0.7656 & 0.4732 \end{pmatrix},$$

with Frobenius fitness $\mathcal{F} = 0.01897$. This equals the theoretical minimum fitness $\mathcal{F}_{\min} = \| |U_{\text{PDG}}| - |U_{\text{PDG}}| \|_F = 0.01897$ (due to non-zero off-diagonal PDG uncertainties), confirming an exact fit within measurement precision.

The Haar-random null test gives $\langle \mathcal{F} \rangle = 0.282$ for Haar-uniform unitaries, with $p = 0.0002$ for the geometric result ($N = 50,000$ trials [3]).

4 Covering tower invariance

4.1 The covering tower of $m003$

The manifold $m003$ sits at the base of a $\mathbb{Z}/5$ -preserving covering tower [4]:

Level	Census index	H_1	vol
Base	1	$\mathbb{Z}/5$	0.9814
Degree-2 cover	39	$\mathbb{Z}/55 \cong \mathbb{Z}/5 \times \mathbb{Z}/11$	1.9627
Degree-3 cover	238	$\mathbb{Z}/80 \cong \mathbb{Z}/16 \times \mathbb{Z}/5$	2.9441

The projection $\mathbb{Z}/55 \twoheadrightarrow \mathbb{Z}/5$ and $\mathbb{Z}/80 \twoheadrightarrow \mathbb{Z}/5$ confirm that the $\mathbb{Z}/5$ torsion is preserved at each level.

4.2 Invariance theorem

Theorem 5 (Covering tower invariance). *Let $p : \tilde{M} \rightarrow M$ be a covering map preserving $H_1 \cong \mathbb{Z}/5$. The Borel fitness and absolute mixing matrix obtained from the optimal gate triple of $\tilde{M}$ are identical to those of M .*

Proof sketch. The covering map p induces $p_* : \pi_1(\tilde{M}) \rightarrow \pi_1(M)$. The holonomy of $\tilde{M}$ factors as $\rho_{\tilde{M}} = \rho_M \circ p_*$. The gate axis $\mathbf{n}(\tilde{\gamma}) = \mathbf{n}(p_*\tilde{\gamma})$ depends only on the image in $\pi_1(M)$. Since the Borel construction depends only on the pairwise dot products $\mathbf{n}_i \cdot \mathbf{n}_j$, and these are determined by the holonomy in $\pi_1(M)$, the resulting mixing matrix is invariant under p . The $\mathbb{Z}/5$ -preservation ensures the flat $U(1)$ CP phase $e^{2\pi i/5}$ is preserved throughout. $\square$

4.3 Numerical verification

Level	Best triple	Fitness	Max element deviation
Base (idx 1)	{aBaB, baab, baaB}	0.01897	—
Deg-2 (idx 39)	{bbb, aBBB, bAAb}	0.01897	$< 10^{-4}$
Deg-3 (idx 238)	{AAb, aaaa, aBAb}	0.01897	$< 10^{-4}$

All three levels produce identical $|U_{\text{geom}}|$ to four decimal places, with maximum element deviation below 10^{-4} . The optimal word triples differ between levels, but produce the same gate configuration up to numerical precision.

5 Homology class no-go theorem

5.1 Flat $U(1)$ holonomy and homology classes

For a compact hyperbolic 3-manifold with $H_1(M) \cong \mathbb{Z}/p$, the flat $U(1)$ connections are classified by $\text{Hom}(\mathbb{Z}/p, U(1))$. The character χ_k assigns to each word $\gamma \in \pi_1(M)$ the holonomy

$$\chi_k(\gamma) = e^{2\pi i k[\gamma]/p},$$

where $[\gamma] \in \mathbb{Z}/p$ is the homology class of γ .

5.2 The no-go theorem

Theorem 6 (Homology class no-go). *Let $\{\gamma_1, \gamma_2, \gamma_3\} \subset \pi_1(M)$ be a word triple with $H_1(M) = \mathbb{Z}/p$. If $[\gamma_i] = [\gamma_j]$ for some $i \neq j$, then the Jarlskog CP-violation invariant $J = 0$ identically for any mixing matrix constructed from this triple via a unitary orthogonalization of the overlap matrix.*

Proof. If $[\gamma_i] = [\gamma_j]$, then $\chi_k(\gamma_i) = \chi_k(\gamma_j)$ for all $k \in \mathbb{Z}/p$. In the 3×3 overlap matrix $O_{mn} = \exp(-\theta_{mn}^2/2\sigma^2)$, the homology-weighted columns i and j are identical: $O_{\cdot i} = O_{\cdot j}$. An overlap matrix with two identical columns has rank ≤ 2 . In the QR decomposition $O = QR$, the matrix Q is unitary and R is upper triangular. When O has rank ≤ 2 , one diagonal entry of R vanishes, so $\det R = 0$ and consequently one column of Q is determined by the other two up to a real phase. Explicitly, Q_{m3} is a real linear combination of Q_{m1} and Q_{m2} , so all 2×2 minors of Q involving column 3 are real. The Jarlskog invariant $J = \text{Im}(U_{11}U_{22}U_{12}^*U_{21}^*)$ is a sum of products of such minors, hence $J = 0$. $\square$

5.3 Application to the quark sector

For $m006 = \text{OrientableClosedCensus}[43]$ with $H_1 = \mathbb{Z}/5$, the optimal CKM word triple has homology classes:

Word	$[\cdot] \in H_1(m006) = \mathbb{Z}/5$
aaB	3
AbA	2
AAb	2

The collision $[\text{AbA}] = [\text{AAb}] = 2$ forces $J = 0$ by Theorem 6. This is confirmed by the identical traces: $\text{Tr}(\rho(\text{AbA})) = \text{Tr}(\rho(\text{AAb})) = -0.1231 + 2.0108i$, placing both words in the same conjugacy class of $\text{SU}(2)$. Verified numerically for 10^3 random same-trace triples ($J = 0$ to machine precision in all cases).

For the lepton sector ($m003$), the optimal PMNS triple has three distinct homology classes; no column degeneracy occurs and $J \neq 0$ is permitted.

Corollary 7 (Quark-lepton CP asymmetry). *The CP suppression in the quark sector ($J_{\text{CKM}} \approx 3 \times 10^{-5}$) relative to the lepton sector ($J_{\text{PMNS}} \approx 0.031$) is a consequence of Theorem 6 applied to the optimal gate triples of $m006$ and $m003$ respectively. It is a topological obstruction, not a numerical coincidence.*

6 Discussion

6.1 Relation to topological quantum computing

In topological quantum computing [1, 2], gates are realized by braiding non-Abelian anyons, with the gate determined by the braid group representation. The protection arises because the gate depends only on the topological class of the braid, not its geometric details.

The covering tower invariance (Theorem 5) provides an analogous form of protection: the PMNS gate configuration depends only on the topological class of the covering, not on which specific words realize the optimal triple at each level. Different words at each level produce the same gate configuration.

This is a different form of protection than braid-based topological quantum computing: it is *discrete* (invariant under covering maps, not continuous deformations) rather than continuous (invariant under isotopy of braids). Whether this discrete protection has implications for physically realizable gate operations is an open problem.

6.2 The no-go theorem as a resource theory

Theorem 6 establishes that homology class collision is an obstruction to CP-violating quantum operations in this framework. This can be reframed as a resource theory: the *homology class diversity* of a gate triple is a resource for producing non-zero J . Triples with all distinct homology classes are the resourceful states; triples with collision are the free states.

This provides a new connection between algebraic topology (the abelianization $\pi_1(M) \rightarrow H_1(M)$) and quantum information theory (the Jarlskog invariant as a measure of CP-violating entanglement capacity).

6.3 What is not claimed

We do not claim that the PMNS mixing matrix arises from a physically realizable quantum circuit acting on a qubit. The construction is mathematical: it produces a unitary matrix from geometric data. Whether any physical qubit system realizes these gates is an open problem requiring identification of a suitable physical platform (e.g., photonic systems with geometric phase, topological insulators, or Chern-Simons anyons).

7 Conclusion

We have established that the holonomy of compact hyperbolic 3-manifolds defines natural qubit rotation gates on the Bloch sphere, and that specific gate configurations arising from the lepton and quark sector manifolds of the hyperbolic flavor geometry framework satisfy two structural theorems.

The covering tower invariance theorem shows that the PMNS gate configuration is a topological invariant of the $\mathbb{Z}/5$ -preserving covering tower of $m003$: identical mixing matrices are obtained from three different manifolds related by covering maps, with maximum element deviation below 10^{-4} .

The homology class no-go theorem provides a topological obstruction to CP-violating quantum operations: gate triples with homology class collision have $J = 0$ identically. The quark sector realizes this obstruction exactly, explaining the observed CP suppression as a topological consequence rather than a numerical coincidence.

These results connect hyperbolic 3-manifold topology, qubit gate theory, and Standard Model flavor physics through the shared mathematical structure of $\mathrm{PSL}(2, \mathbb{C})$ holonomy representations.

Data Availability

All scripts are available at <https://github.com/drmlgentry/hyperbolic-flavor-scan>. The scripts `qubit_gates_v2.py` and `borel_covers_tower.py` reproduce all numerical results.

Conflict of Interest

The author declares no conflicts of interest.

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